

Question 31 (a)

Criteria	Marks
- Explains all the graphs with respect to the negative feedback control of blood glucose in healthy humans - Makes clear references to features of the graphs - Shows detailed understanding of control of blood glucose	6
- Explains the graphs with respect to the negative feedback control of blood glucose in healthy humans - Makes references to features of the graphs - Shows understanding of control of blood glucose	5
- Describes the graphs - Links the graphs to features of the negative feedback control of blood glucose	4
- Refers to feature(s) of the graphs - Links the feature(s) to the negative feedback control of blood glucose	3
- Refers to features of the graphs OR - Refers to a feature of a graph - Provides a feature of negative feedback control of blood glucose	2
Provides some relevant information	1

Sample answer:

The plasma levels measured in the first 60 minutes represent resting levels. After the meal, plasma glucose rises as a direct result of absorption of glucose from the gut into the bloodstream.

Rising blood glucose stimulates β cells in the pancreas to release the hormone insulin, which stimulates body cells to take up glucose to be used in their metabolism and the liver to take up glucose to be stored as glycogen. Therefore, the rise in insulin in Figure 2 is a direct result of the rise in plasma glucose. As the cells and liver take up glucose, they remove the glucose from the blood leading to the subsequent fall in plasma glucose levels. The falling glucose removes the stimulation of β cells and by this negative feedback mechanism, insulin levels also fall.

Glucagon is a hormone that is an important part of the negative feedback loop that controls glucose levels in the blood. If plasma glucose falls to low levels, glucagon is released from alpha cells in the pancreas and causes glucose to be released into the blood from glycogen stores in the liver and muscles to restore normal glucose levels. In Figure 1, as glucose rises the alpha cells will produce less glucagon and falling glucagon levels can be seen in Figure 3. This will result in less glucose being released from the liver, reducing glucose levels.

Question 31 (b)

Criteria	Marks
• Outlines differences between control of blood glucose and body temperature	3
• Outlines a difference between control of blood glucose and body temperature OR • Identifies two differences	2
• Provides some relevant information	1

Sample answer:

Temperature changes are detected by the hypothalamus in the brain but changes in glucose are detected by the pancreas. The response to changes in temperature is via the nervous system but glucose is regulated via hormones.